

II. World Geography

The Universe

i. General Concept

*The first rational perspective regarding the origin of the planet was probably proposed by a French scientist 'Comte de Buffon' in 1749. According to the number of stars participating in the origin of planets, the scientific concepts have been divided into two parts- (1) **Monistic concept** and (2) **Dualistic concept**.

*Under the Monistic concept, Immanuel Kant's Gaseous Hypothesis and Pierre Simon Laplace's Nebular Hypothesis became famous. Under the Dualistic concept, Chamberlin and **Moulton** Planetesimal Hypothesis, James Jeans and Jeffery's Tidal Hypothesis, H.N. Russell and Littleton Binary Star Hypothesis etc. became noteworthy.

*The latest theory regarding the origin of the universe is the **Big Bang Theory**. ***George Lemaitre** (1894-1966) was the first to present the idea of an expanding universe. Later in 1967, Robert Wagoner presented the interpretation of this theory. The Big Bang theory explains how **13.7** or **13.8** billion years ago universe expanded from a very high density and high-temperature state. *After the explosion, the universe expanded in all directions and different galaxies of different shapes were formed. Thus, the universe comprises of an infinite number of galaxies.

It is estimated that there are appx. 100-200 billion galaxies in the universe and every galaxy is estimated to have **around 100 billion stars**. Our galaxy is known as an **Akash Ganga** or **Milky Way**.

*The shape of the Galaxies varies in shape and size. Some are spiral, some are irregular and some are elliptical. *In 2005, the Spitzer Space Telescope gave clear evidences that our galaxy (the Milky Way) is **barred spiral** in shape, hence disapproving the earlier theories. *The universe is a collection of infinite galaxies. Every galaxy has around 100 billion stars. These stars are formed as a result of the presence of hydrogen and helium gases in their unified form (Nebulae). The gravitational force keeps the gases and dust particles in the unified form. Velocity, high temperature and organizational processes are the factors behind the formation of a star. *The formation of the star passes through different phases. In the first phase, a **proto-star** is formed after that the star enters the second phase i.e., the **main sequence Star**.

A star of the size of our sun stays in this phase for around 9 or 10 billion years. After this phase, the fuel of the star i.e., hydrogen starts getting used up to an extent that the star enters the third phase i.e., towards its **death**. *The lifespan of any star depends upon its size. A star of the size of our sun (one solar mass) has a lifespan of around 9 or 10 billion years. The bigger the star is, the shorter is its life span. For example, a star whose surface temperature is more than 30000 Kelvin and which is 100 times the size of our sun has a life span of only a few million years. After the main sequence life of the star ends, it enters in the elderly phase.

*In this phase, the outer layer of the star expands greatly, It becomes a **Red Giant** or **Red Super giant**. *After the red Giant or super Red Giant phase, a star witnesses an explosion in the form of a **Nova** or **Supernova**, thus entering its last stage of life. Depending upon its size it can end up in- a **Black Dwarf**, **Neutron star** or a **Black Hole**.

Sun like small stars – after the Red Giant phase and Nova explosion if the stellar mass is less than 1.44 solar Masses, the star will end up being a White Dwarf first and then into a Black Dwarf.

Medium sized stars - after the Red Giant Phase and Supernova explosion, if the stellar mass is between 1.44 to 3 solar masses, then a star can convert into a Neutron Star (Pulsar).

Large size stars – those which have a stellar mass more than 3 times that of our sun. These stars have the possibility to convert into a black hole after a supernova explosion. *In the year 1931 A.D. Indo-American Physicist **Subrahmanyan Chandrasekhar** invented the Chandrasekhar Limit. The Chandrasekhar limit which is appx. **1.44** times the mass of the sun (1.4 solar mass) determines the death state of a star. Any white dwarf with less than 1.4 solar mass will remain a white dwarf forever, while a star that exceeds this mass is destined to end in life in the most violent of explosions: a supernova, creating neutron stars or black holes. *It is to be noted here that the Chandrasekhar limit is determined after the nova or supernova explosion of the star has taken place. Prima facie it is determined by the mass of the residue of the exploded star. *It is not certain that which of the main sequence stars leave a residual mass under or over 1.44 times of solar mass, after explosion. However it is estimated that stars, with 8 times the size of our sun leave a residue

equal to 1.44 times of solar mass after the explosion, and stars bigger than the above ones leave a residual mass of more than 1.44 times the solar mass. The stars of the latter case have the probability of creating **black holes**. The high gravitational force of the black hole sucks in any left residue of the exploded star. The gravitational pull is so strong that it does not even spare passing by objects; with the passing of time the black hole gets smaller in size, until it reaches the size of a grain of sand.

*Black hole related information was presented by Dr. S. Chandrasekhar on 11th January 1935 at Royal Astronomical Society of London. He presented the original papers regarding the subject. In the year 1983, a combined Nobel Prize in Physics was awarded to Dr. S. Chandrasekhar and W.A. Fowler. After a supernova explosion, the star becomes dead. Hence, a supernova is the explosion of a dying star.

*The **sun** is one of the stars of our Galaxy, the Milky Way. Its distance from the Galactic center is **7.94 kiloparsecs** or 25,896 thousand light years. *With an average speed of 720 thousand km/hour, it takes the sun approximately 22.5 to 25 crore years for one revolution of the Galaxy (Every star in the Galaxy revolves around the Galactic Center). *Hence the duration of the time required for the sun to orbit once around the Galactic center of the Milky Way Galaxy is known as 1 **cosmic year** or 1 **Galactic year**. *A **light year** is a unit of length in astronomy and is equivalent to the distance traveled by light in one year in vacuum (appx. speed of light in vacuum 3×10^8 km/sec). Light years is the unit used to measure the distance between the stars. The distance traveled in one light year in the different units is as follows -

In metre 9.461×10^{15}

In kilometre 9.461×10^{12}

Millimetre 9.461×10^{18}

Astronomical unit 63.24×10^3

Miles 5.878×10^{12}

***Chandrasekhar limit** is a limit exceeding which, the free revolving electrons get energized enough to leave their orbit, leaving behind the neutrons. This condition is known as a **neutron star** or **pulsar**. A group of stars that forms an imaginary outline or pattern on the celestial sphere, typically representing an animal, mythological person or creature, God, or an inanimate object is known as a **constellation**. A general list of **88** constellations was produced for the International Astronomical Union in 1922. *The Great Bear constellation is also called **Ursa Major**, **Big Bear** and **Saptarishi**. With the help of great bear constellation, the position of the pole star can be easily located. It is situated immediately above

the North Pole of the Earth.

*The Orion Nebula is a diffuse nebula situated in the Milky Way. It is one of the brightest nebulae and is visible to the naked eye in the night sky.

***Proxima Centauri** is the closest star to our Sun. Betelgeuse centauri is the second- brightest star in the constellation. It is around 700 light years from Earth. The Hubble Telescope took the first picture of it. After the sun it was the second star whose picture was taken. Stars revolve around the North and South celestial poles, and parallel to the celestial equator.

*The north and south celestial poles appear directly overhead the earth's North Pole and south pole respectively likewise the celestial equator is placed directly overhead the earth's equator and always stays perpendicular (90°) to the poles. The celestial equator is defined to be infinitely distant. Thus, the ends of the semi-circle always intersect the horizon due east and due west. Regardless of the observer's position on earth. *A man sitting in a space shuttle can see the sky only black because in space there is no atmosphere.

1. The "Big Bang Theory" is related to –

- (a) Continental Drifts (b) Origin of the Universe
- (c) Origin of the Himalayas (d) Eruption of Volcanoes

R.A.S./R.T.S.(Pre) 2007

Ans. (b)

The "Big Bang Theory" is related to the origin of the Universe, Milky Way and Solar System. This theory was propounded by George Lemaitre (1894-1966). Later in 1967, Robert Wagner explained this theory. According to this theory, the whole universe was concentrated in a hot and dense point some 13.7 or 13.8 billion years ago. Due to high concentration, there was a sudden explosion which was termed the Big-Bang.

2. The "Big-Bang Theory" explains the origin of –

- (a) Mammals (b) Ice-age
- (c) Universe (d) Ocean

U.P.P.C.S. (Pre) 2016

Ans. (c)

The "Big Bang Theory" explains the origin of the universe from the earliest known period through its subsequent large scale evolution. The measurement of the expansion rate of the universe places this moment approximately 13.7 or 13.8 billion years ago, which is thus considered the age of the universe.

3. The theory of continental drift was developed by-

- (a) J.J. Wilson (b) A. Wegener

- (c) Du Toit (d) H. Hess
(e) None of the above / More than one of the above.

67th B.P.S.C. (Pre) 2022

Ans. (b)

The theory of continental drift is most associated with the German geophysicist Alfred Wegener.

4. The milky way is classified as –

- (a) Spiral Galaxy
(b) Electrical Galaxy
(c) Irregular Galaxy
(d) Round Galaxy

U.P.P.C.S. (Pre) 2001

Ans. (a)

According to the theory propounded by Georges Lemaitre, 13.7 or 13.8 billion years ago the total cosmic substance was in the form of dense mass which spread to form a number of galaxies. An infinite group of these swirling galaxies is visible to us collectively in the form of the universe. Every galaxy is estimated to contain around 100 billion stars. The galaxy in which our solar system is situated is known as the 'Milky Way'. There are various shapes of galaxies, for example elliptical, spiral, irregular etc. Spitzer Space Telescope in 2005, produced clear proof against the preconception about the shape of the Milky Way and made it clear that the shape of our Milky Way is barred spiral.

5. Time taken by the Sun to revolve around the centre of our galaxy is –

- (a) 2.5 crores years (b) 10 crores years
(c) 25 crores years (d) 50 crores years

40th B.P.S.C. (Pre) 1995

Ans. (c)

The Sun is one of the stars of our galaxy (Milky Way). Every star of the galaxy revolves around the galactic centre and the time taken by a star in revolving around this centre depends upon the distance between the galactic centre and the star. Sun is relatively far from the centre, so it takes more time to revolve around it. The Sun is at a 7.94 KPC or 25,896 thousand light years away from the centre and with a speed of 720 thousand km/hour, it takes 22.5 to 25 crore years, to revolve around the centre of the Milky Way galaxy. This period is called 'Cosmic Year or Galactic Year'.

6. The colour of the star indicates its –

- (a) Distance from Sun (b) Lighting or glow
(c) Distance from Earth (d) Temperature

I.A.S. (Pre) 1994

Ans. (d)

The colour of a star indicates its temperature. The temperature of stars of different colours is as follows –

(Colours)		(Temperature range)
Blue	-	50,000-28,000 K
Blue-white	-	28,000-10,000 K
White	-	10,000-7500 K
White Yellow	-	7500-6000 K
Yellow	-	6000-4900 K
Orange	-	4900-3500 K
Red	-	3500-2000 K

7. The limit beyond which stars suffer internal collapse is called :

- (a) Chandrasekhar limit
(b) Eddington limit
(c) Hoyle limit
(d) Fowler limit

U.P.P.C.S. (Pre) 1997

Ans. (a)

Subrahmanyan Chandrasekhar was an Indo-American physicist who described the "Chandrasekhar Limit". This limit describes the maximum mass of a white dwarf star or the mass limit above which a star will ultimately collapse into a neutron star or a black hole. He revealed that this mass was approximately 1.44 times of the solar mass. In 1983 Chandrasekhar was awarded the Nobel Prize in Physics for his work on the evolution of stars.

8. Black-hole is :

- (a) a flight recorder in aeroplane
(b) a spot on the sun
(c) a place in Antarctica
(d) a collapsed star

U.P.P.C.S. (Pre) 2019

Ans. (d)

According to the American space agency NASA, most black holes form the remnants of a large star that dies in a supernova explosion.

9. Which is a celestial phenomenon occurring due to stars?

- (a) Ozone hole (b) Black Hole
(c) Rainbow (d) Comet

U.P.P.C.S. (Pre) 1997

Ans. (b)

See the explanation of the above question.

10. The "Black Hole Theory" was propounded by :

- (a) C.V. Raman (b) H.J. Bhabha
(c) S.Chandrashekhar (d) Hargovind Khurana

U.P.P.C.S. (Pre) 1996

Ans. (c)

See the explanation of the above question.

11. The 'Black Hole' was first made known by

- (a) Herman Bandy
(b) Megh Nath Saha
(c) S. Chandrasekhar
(d) J.V. Narlikar

U.P.P.C.S. (Pre) 2015

Ans. (c)

Subrahmanyan Chandrasekhar was a great scientist and Nobel Prize winner who gave details on black hole theory. He is known for the Chandrasekhar limit which showed that there is a maximum mass that is supported against gravity by pressure made up of electrons and atomic nuclei (the value of this limit is about 1.44 times of the solar mass) which was derived by Chandrasekhar in 1931. He was awarded Nobel Prize for Physics jointly with W.A. Fowler in 1983.

12. Information about 'Black Hole' was first provided by –

- (a) S. Chandrashekhar (b) Hermann Bondi
(c) Rutherford (d) Copernicus

U.P.P.C.S. (Mains) 2016

Ans. (a)

From the given options the information about 'Black Hole' was presented by Dr. S. Chandrashekhar. On 11 January, 1935 he presented a research paper in the Royal Astronomical Society of London and propounded that a white dwarf star turns into a Black Hole after attaining a certain mass. He was jointly awarded the Nobel Prize for Physics in 1983 with W.A. Fowler.

13. Assertion (A) : A black hole is an astronomical entity that cannot be seen by telescope.

Reason (R) : The gravitational field on a black hole is so strong that it does not allow even light to escape.

In the context of the above statements, which one of the following is correct :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but R is false.

(d) (A) is false, but R is true.

U.P.P.C.S.(Pre) 2003

U.P.P.C.S.(Pre) 2001

Ans. (a)

Assertion (A) and Reason (R) both are true and Reason (R) explains Assertion (A) properly. It is not possible to observe a black hole directly with the help of a telescope. The intense gravitational field of a black hole due to the high density does not allow any type of radiation. Even light rays cannot escape from this. The presence of black hole is ascertained by the behaviour of stars, gases and masses near it with the help of space telescope.

14. A 'black hole' is a body in space which does not allow any type of radiation to come out. This property is due to its –

- (a) Very small size (b) Very large size
(c) High density (d) Very low density

I.A.S. (Pre) 2000

Ans. (c)

See the explanation of the above question.

15. A 'Supernova' is –

- (a) An asteroid (b) A black hole
(c) A comet (d) A dying star

U.P. Lower Sub. (Mains) 2015

Ans. (d)

A supernova is a large explosion that takes place at the end of a star's life cycle. A supernova happens where there is a change in the core or center of a star. A change can occur in two different ways, with both resulting in a supernova. The first type of supernova happens in binary star systems wherein two stars orbit the same point. The second type of supernova occurs at the end of the a single star's lifetime.

16. How many constellations are in our space?

- (a) 87 (b) 88
(c) 89 (d) 90

44th B.P.S.C. (Pre) 2000

Ans. (b)

According to the International Astronomical Union, there are 88 constellations in the sky. Most of these imaginary patterns can be seen from the southern hemisphere. It will take a full year to get a glimpse of all the constellations.

17. Which one of the following is not an Astronomical object?

- (a) Pulsar (b) Brittle star
(c) Black hole (d) Quasar

I.A.S. (Pre) 1993

Ans. (b)

Brittle star is not an astronomical object. Brittle stars are echinoderms in the class Ophiuroidea closely related to starfish. These are aquatic creatures and crawl on the sea floor, while a Pulsar is a celestial body emitting very regular pulses of radio waves. A quasar is an astronomical object of very high luminosity found in the centers of some galaxies at a distance of billions of light-years. A black hole is also a cosmic body of extremely intense gravity from which nothing (even light) can escape and is formed by the death of a massive star. Thus, option (b) is the correct answer.

18. Which one of the following is NOT found in space?

- (a) Pulsar (b) Brittle Star
(c) Black Hole (d) Quasar

U.P.P.C.S. (Pre) 1996

Ans. (b)

Pulsar, Black Hole and Quasar are found in space while Brittle Star is an echinoderm closely related to Starfish. Hence Brittle Star is related to the sea, not space.

19. A group of stars arranged in a particular shape, is called–

- (a) Milky Way (b) Constellation
(c) Andromeda (d) Solar system

U.P.P.C.S. (Pre) 2013

Ans. (b)

A constellation is a group of stars forming a recognizable pattern. It is traditionally named after its apparent form or identified with a mythological figure. Hence, option (b) is the correct answer.

20. Hubble space telescope has captured, for the first time, detailed image of the surface of a distant star. Name of the star :

- (a) Virgo (b) 70 Virgins
(c) Betelgeuse (d) Big Dipper

U.P.P.C.S. (Pre) 1995

Ans. (c)

Betelgeuse is the second-brightest star in the constellation of Orion. It is at a distance of about 700 light years from the Earth. It was the second star after the Sun, whose image had been captured.

21. Consider the following Statements –

A person in spacecraft situated at the mid of earth and Sun will see that –

1. Sky is black
2. Stars do not twinkle
3. The temperature outside the spacecraft is more than that of the earth surface.

Which of the following is true ?

- (a) Only 3 (b) 1 and 2
(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 1996

Ans. (b)

Due to the absence of atmosphere in space, the sky appears black and stars do not twinkle. Twinkling of stars is caused by the passing of light through different layers of turbulent atmosphere similarly it is due to the atmosphere that light scatters to let us see the blue sky. Outside the spacecraft (in space) the temperature is 2.7 Kelvin (-270.45°C) which is much less than the temperature of the Earth's surface. So, statement 3 is wrong.

22. "Light Year" is the unit of -

- (a) Time (b) Distance
(c) Luminosity (d) None of these

M.P.P.C.S. (Pre) 1994

Ans. (b)

Light year is the distance which light can travel in a vacuum in one year. In different units of distance, one light year is equivalent to as follows-

Metre 9.461×10^{15}
Kilometre 9.461×10^{12}
Millimetre 9.461×10^{18}
Celestial unit 63.24×10^3
Miles 5.878×10^{12}

23. The unit of measuring distance between the stars is known as

- (a) Stellar mile (b) Cosmic km.
(c) Galactic unit (d) Light year

Uttarakhand P.C.S. (Pre) 2006

Ans. (d)

Light year is most often used in measuring the distance between the stars. The distance covered by light in one year in a vacuum with a speed of 3×10^5 km./sec is known as 'Light Year'.